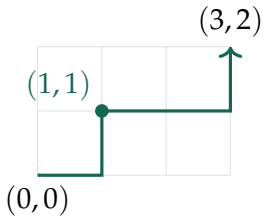
Paths, combinations and Pascal

High school · counting · Lesson and method · 05

Prerequisites. Products, factorials and unordered choices.

Key point · Ideas and method. A path from $(0,0)$ to (a,b), using only rightward or upward steps, has $a+b$ steps. Choosing the positions of the a horizontal steps gives $\binom{a+b}{a}$ paths.
For $0 \leq k \leq n$, $\binom{n}{k} = \frac{n!}{k!(n-k)!}$. The order of the chosen objects does not matter. Symmetry $\binom{n}{k} = \binom{n}{n-k}$ corresponds to choosing included or excluded objects.
Pascal's identity $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$ splits choices by whether one fixed object is included. Paths can also be classified by their last step.

Example 1. From $(0,0)$ to $(3,2)$, there are $\binom{5}{3} = 10$ paths. To pass through $(1,1)$, choose one of two paths to that point and one of three paths onward: $2 \times 3 = 6$ paths.



Activity 1 · Explore. Draw a small grid and write the number of arriving paths at each vertex. Add the values of its left and lower neighbours.